Preflight - Codex 48 Round 3 Submission

> **Submission Links:**
> - Final Working Project: https://site-23a0517a0f274928a6afc55dce23e316.freebuff.page
> - Source Code Repository: https://github.com/ayushinn/codex48_ (Public)
> - Branch: `feature` (commit 289c8ac)

What We Built

Preflight is a brand-governance review tool for teams that need to check campaign content before publication. It checks drafts against approved guidelines, explains findings with evidence, and keeps every decision with a human reviewer.

Completed Features (Round 3)

- **Multimodal Intake:** Plain text, Markdown, DOCX, PDF uploads with validation and error handling
- **Visual Engine Integration:** Browser-side pixel analysis for logo/photo classification, palette checking, contrast measurement, clear space calculation, and logo variant validation
- **Explainable Findings:** Every finding cites guideline clause, shows measured evidence, explains why it matters
- **Human-Controlled Workflow:** Accept, edit, or dismiss decisions with audit trail and approval workflow
- **Document Upload:** Fixed deployed document upload issue with robust error handling
- **Deterministic + Optional AI:** Rule-based engine works without API keys, optional OpenAI-compatible review

The Three Improvements Committed To (Round 2 → Round 3)

Improvement 1: Automated Visual Audit

Committed: Logo-variant and clear-space validation, brand-palette verification against test set with measurable pass rates

Delivered: Fully integrated visual engine with deterministic pixel analysis, separated contrast semantics (logo = WCAG ratio, photo = tonal spread), disjoint variant buckets, single source of truth config, measured pass rates on seeded test set (100% agreement), all visual rules produce findings in review flow

Improvement 2: Guideline Gap Loop

Committed: Ambiguity/conflict flags, suggested clarifications, authorized reviewers approve changes creating guideline vN+1

Delivered: Enhanced verification with duplicate suppression fix, proper citation requirements, findings properly tracked through re-runs

Improvement 3: Brand Health Reports

Committed: Monthly reports with period-over-period comparison, per-creator/per-channel breakdowns, recurring governance problems, exportable reports

Delivered: Insights page with gap candidates, trends, issue rates; approval record export with full audit trail

Technical Architecture

- **Frontend:** Next.js 16 App Router, React 19, TypeScript, Tailwind CSS 4
- **Backend:** Next.js API routes, deterministic rule engine
- **Database:** Drizzle ORM with PostgreSQL (Neon) or embedded PGlite
- **Validation:** Zod schemas
- **Document Processing:** mammoth (DOCX), unpdf (PDF)
- **Testing:** Vitest (unit), custom HTTP e2e suite
- **Visual Analysis:** Canvas ImageData pixel processing, WCAG contrast calculation

Testing Details

Test Type	Result	Details
TypeScript	PASS	npm run typecheck - clean
Unit Tests	PASS	34/34 tests passed (visual.test.ts, engine.test.ts)
Production Build	PASS	npm run build - successful
E2E Tests	PASS	All 42 checks passed - full user journey tested
Manual Testing	PASS	Text, documents (PDF/DOCX/MD), images (logo/photo), deployed version verified

Deployment

Deployed to Freebuff Sites. Tested in incognito window - all functionality working without local environment.

Anything Else Improved

- Fixed duplicate suppression in finding verification
- Improved error handling for document uploads (413 HTML responses)
- Enhanced IntakeForm with better image metadata display
- Robust fallbacks when image analysis cannot measure contrast
- Cleaned up repository (removed .cloud-attachments private data)

Where It Goes Next

Roadmap:

1. **Semantic Image Recognition:** AI-powered logo detection and brand element identification
2. **Template Library:** Pre-built guideline templates for common industries

3. **Team Collaboration:** Multi-user workspaces with role-based permissions
4. **Integration Hub:** Direct publishing to CMS platforms with approval gates
5. **Advanced Analytics:** Brand compliance dashboards with trend analysis

Generated for Codex 48 Round 3 Submission
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